

Figure 1: Three-layer hierarchical architecture for adaptive shared-autonomy assistive navigation.

L1: Preference-RL (1 Hz)

- Input: user trajectory segments (50 steps)
- Algorithm: CQL (offline) + MaxEnt IRL (online)
- Output: cost weights $w \in \mathbb{R}^m$
- Role: learns user preferences from behavior

L2: Tube MPC (10 Hz)

- Input: state $\hat{x}_0$, cost weights w , obstacle map
- Solver: CasADi NLP $\rightarrow$ qpOASES QP
- Output: nominal trajectory $\tilde{x}$ + tube Ω
- Guarantee: Proposition 1 ($x \notin C, \forall w \in W_{\text{bdd}}$)
- Fallback: Minimal Risk Condition (MRC)

SAFETY BOUNDARY
(L2 only, never exposed to L1)

L3: DWA (100 Hz)

- Input: tube X_k , dynamic obstacle detections
- Search space: $U \oplus K\Omega$ (constrained by tube)
- Output: $(v_{\text{cmd}}, \omega_{\text{cmd}})$ via vibrotactile + audio UI
- Emergency: brake + trigger L2 replan

↑
*cost function
parameters*

↑
*reference trajectory
+ tube cross-section*

↓
*guidance
($v_{\text{cmd}}, \omega_{\text{cmd}}$)*